

KLE Technological University
Belagavi

A Project Report on

“Smart Grid Anomaly Detection using Spectrogram Analysis and Attention-Based Deep Learning”

A Project Report Submitted in Partial Fulfillment of the Requirement for the Course of
Deep Learning (24ECAC305)

in

6th Semester of Computer Science(Artificial Intelligence) and Engineering

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DECLARATION

We hereby declare that the matter embodied in this report entitled “**Smart Grid Anomaly Detection using Spectrogram Analysis and Attention-Based Deep Learning**” submitted to KLE Technological University for the course completion of Deep Learning (24ECAC305) in the 6th Semester of Computer Science and Engineering is the result of the work done by us under the guidance of Dr. Salma Shahapur.

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CERTIFICATE

This is to certify that the project entitled “**Smart Grid Anomaly Detection using Spectrogram Analysis and Attention-Based Deep Learning**” submitted to KLE Technological University’s Dr. MSSCET, Belagavi for the partial fulfillment of the requirement for the course Deep Learning (24ECAC305) is a bonafide record of the work carried out by the students under my supervision.

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Abstract

Smart grids generate large amounts of real-time electrical data such as voltage, current, power consumption, and frequency measurements. Detecting abnormal conditions in these systems is important because faults such as overloads, transformer failures, harmonic distortion, and frequency instability can lead to equipment damage and power outages. Traditional monitoring systems mainly depend on manual supervision and threshold-based methods, which are often unable to detect complex anomalies efficiently.

This project proposes a Deep Learning-based Smart Grid Anomaly Detection system using spectrogram analysis and transfer learning techniques. Raw smart grid time-series signals are converted into spectrogram images using Short-Time Fourier Transform (STFT), which provides a time-frequency representation of the signals. These spectrograms help visualize abnormal frequency patterns and disturbances clearly.

Pretrained Convolutional Neural Network (CNN) models such as ResNet, DenseNet, and EfficientNet were used for anomaly classification through transfer learning. To address severe dataset imbalance, optimization techniques such as Focal Loss, WeightedRandomSampler, Mixup augmentation, and learning rate scheduling were applied. In addition, a custom attention-based architecture called GAResNet was developed to improve anomaly localization and interpretability.

Experimental results demonstrated that the proposed system achieved improved anomaly detection performance and higher sensitivity toward abnormal grid conditions. The project shows that combining spectrogram analysis, transfer learning, and attention mechanisms can provide an effective and explainable solution for intelligent smart grid monitoring.

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Chapter 1

Introduction

1.1 Introduction to Smart Grid

A smart grid is an advanced electrical power system that uses digital communication technologies, sensors, automation, and intelligent monitoring systems to improve the efficiency, reliability, and stability of electricity distribution. Unlike traditional power grids, smart grids continuously collect and analyze real-time data from different components such as transformers, substations, smart meters, and renewable energy systems.

Modern power systems are becoming more complex due to increasing electricity demand and the integration of renewable energy sources such as solar and wind power. Because of this complexity, maintaining grid stability and power quality has become a major challenge. Any abnormal condition in the grid, such as overloads, harmonic distortion, transformer faults, or sudden frequency changes, can affect the entire electrical network and may lead to equipment damage or power outages.

To maintain reliable operation, modern smart grids require intelligent monitoring systems capable of detecting anomalies automatically and in real time.

1.2 Need for Smart Grid Anomaly Detection

Power grid anomalies are unexpected disturbances or abnormal operating conditions that occur within the electrical network. These anomalies may include voltage fluctuations, harmonic distortion, transformer failures, transient spikes, overload conditions, or frequency instability.

Traditional monitoring systems mainly rely on manual supervision and threshold-based alarm systems. Human operators can monitor only a limited number of measurement points at a time, while modern smart grids generate huge volumes of real-time data from thousands of sensors. Conventional threshold-based methods are often unable to detect complex fault patterns and hidden anomalies efficiently.

If anomalies are not detected at an early stage, they may propagate across the grid and cause cascading failures, equipment damage, or large-scale blackouts. Therefore, intelligent AI-based anomaly detection systems are essential for ensuring reliable and uninterrupted electricity supply.

1.3 Role of Deep Learning in Smart Grids

Deep Learning is a branch of Artificial Intelligence that enables computers to learn complex patterns directly from data. Deep learning models, especially Convolutional Neural Networks (CNNs), have shown excellent performance in image classification, signal processing, and anomaly detection applications.

In this project, smart grid time-series signals are converted into spectrogram images. A spectrogram is a visual representation of signal frequencies over time. Deep learning models can analyze these spectrograms and learn the difference between normal and abnormal grid behavior automatically.

Transfer learning techniques using pretrained CNN models such as ResNet18, DenseNet121, and EfficientNet-B0 were used to improve anomaly detection performance.

1.4 Problem Statement

Traditional power grid monitoring systems face several limitations in handling large-scale real-time data. Manual monitoring is time-consuming and cannot effectively identify hidden or complex anomalies. Existing threshold-based systems also fail to capture transient disturbances and frequency-domain abnormalities accurately.

Another major challenge is severe dataset imbalance, where normal operating conditions are far more common than anomaly conditions. Due to this imbalance, many machine learning models become biased toward predicting normal conditions and fail to detect rare anomalies effectively.

Hence, there is a need for an intelligent anomaly detection framework that can process smart grid signals, identify abnormal patterns automatically, and provide reliable monitoring for modern electrical networks.

1.5 Objectives of the Project

The main objectives of this project are:

- To convert smart grid time-series signals into spectrogram images using Short-Time Fourier Transform (STFT).
- To apply deep learning techniques for smart grid anomaly detection.
- To compare the performance of different transfer learning models.
- To handle severe class imbalance using advanced optimization techniques.
- To improve anomaly localization using attention mechanisms.
- To develop an intelligent and explainable smart grid monitoring system.

1.6 Proposed Methodology

The proposed system first collects smart grid operational data such as voltage, current, and power consumption signals. The raw time-series signals are then preprocessed and divided into smaller windows using a sliding window technique.

After preprocessing, Short-Time Fourier Transform (STFT) is applied to convert the signals into spectrogram images. These spectrograms provide a time-frequency representation where anomalies become visually distinguishable.

The generated spectrogram images are used to train deep learning models such as ResNet18, DenseNet121, and EfficientNet-B0. Advanced optimization techniques including Focal Loss, WeightedRandomSampler, Mixup augmentation, and learning rate scheduling are used to improve performance under severe class imbalance conditions.

A custom attention-based architecture called GAResNet is also implemented using Spatial Attention Modules (SAM) and Channel Attention Modules (CAM) to improve anomaly localization and model interpretability.

1.7 Scope of the Project

The scope of this project includes intelligent monitoring and anomaly detection in smart power grids using deep learning and spectrogram analysis. The proposed system can automatically analyze electrical signals and classify them as normal or anomalous.

The system can be extended for real-time deployment in substations, renewable energy systems, industrial monitoring systems, and smart city infrastructures. Future integration with IoT devices, cloud computing, and edge AI can further improve scalability and operational efficiency.

1.8 Advantages of the Proposed System

The proposed smart grid anomaly detection system provides several advantages:

- Automatic real-time monitoring of smart grid conditions.
- Early detection of faults and abnormal conditions.
- Reduced dependency on manual supervision.
- Improved reliability and stability of power systems.
- Better anomaly localization using attention mechanisms.
- Explainable AI-based decision-making using spectrogram analysis.

Chapter 2

Literature Survey

Existing intelligent transportation systems demonstrate the effectiveness of GPS-based tracking, cloud computing, and data analytics in improving public transport efficiency and commuter experience.

2.1 Research Gap

- Over-reliance on dedicated GPS hardware increasing deployment cost.
- Lack of continuous real-time tracking in RFID systems.
- Limited integration between commuter, driver, and admin platforms.
- Absence of scalable solutions for resource-constrained environments.

Chapter 3

Methodology

3.1 Introduction

The methodology of the proposed Smart Grid Anomaly Detection system involves multiple stages including data collection, preprocessing, spectrogram generation, deep learning model training, optimization, and evaluation. The primary objective of the system is to identify abnormal operating conditions in smart grids using spectrogram-based deep learning techniques. The overall workflow converts raw time-series electrical signals into spectrogram images and classifies them into normal or anomaly categories using advanced Convolutional Neural Network (CNN) architectures.

3.2 Overall System Architecture

The proposed system consists of the following major stages:

1. Data Collection
2. Data Preprocessing
3. Sliding Window Segmentation
4. Spectrogram Generation using STFT
5. Deep Learning Model Training
6. Optimization and Class Balancing
7. Anomaly Classification
8. Performance Evaluation

Diagram to Add: Overall System Architecture Diagram showing:

Dataset → Preprocessing → STFT Spectrogram Generation → CNN Models →
Attention Mechanism → Classification Output

3.3 Dataset Collection

The dataset used in this project consists of smart grid operational measurements collected from electrical monitoring systems. The dataset contains multiple parameters related to power system behavior such as voltage, current, power consumption, reactive power, power factor, solar power, wind power, temperature, and humidity.

The primary signal selected for spectrogram generation is the **power consumption** signal because it captures important variations related to abnormal grid behavior.

3.3.1 Dataset Characteristics

Parameter	Value
Total Samples	663
Training Samples	545
Testing Samples	118
Normal Samples	83%
Anomaly Samples	16%
Window Size	128
Step Size	64

Table 3.1: Dataset Characteristics

3.4 Data Preprocessing

Data preprocessing is an important step to improve the quality of input data before model training. The raw smart grid dataset may contain missing values, inconsistent measurements, and noise.

The following preprocessing techniques were applied:

- Missing values were handled using forward-fill and zero-fill methods.
- Signals were normalized to improve training stability.
- Sliding window segmentation was applied to divide continuous signals into smaller fixed-size windows.
- Each segmented window was labeled as normal or anomaly depending on the percentage of abnormal samples present.

Diagram to Add: Data Preprocessing Pipeline Diagram.

3.5 Sliding Window Technique

The continuous time-series signals were divided into smaller overlapping windows using a sliding window approach. This technique helps capture local temporal variations within the electrical signals.

The following parameters were used:

- Window Size = 128
- Step Size = 64

Each window represents a short-duration electrical signal segment that is later converted into a spectrogram image.

Diagram to Add: Sliding Window Illustration showing overlapping windows.

3.6 Spectrogram Generation using STFT

Short-Time Fourier Transform (STFT) was used to convert one-dimensional electrical signals into two-dimensional spectrogram images.

STFT divides the signal into smaller segments and applies Fourier Transform on each segment separately. This allows the system to analyze how signal frequencies change over time.

The generated spectrogram contains:

- X-axis: Time
- Y-axis: Frequency
- Color Intensity: Signal Strength

Normal grid conditions usually show stable 50 Hz frequency patterns and clean harmonics, while anomalies appear as frequency shifts, spikes, harmonic distortions, or broadband noise.

Logarithmic normalization using `log1p()` was applied to improve visibility of low-power frequency components.

3.6.1 STFT Parameters

Parameter	Value
Method	scipy.signal.spectrogram
nperseg	64
noverlap	32
Image Size	224×224
Colormap	Viridis
DPI	100

Table 3.2: STFT Spectrogram Parameters

Diagram to Add:

- STFT Working Diagram
- Example Spectrogram Image (Normal vs Anomaly)

3.7 Deep Learning Models

Transfer learning-based CNN architectures were used for anomaly classification. Pre-trained ImageNet weights were utilized to improve feature extraction capability and reduce training time.

The following models were implemented:

3.7.1 ResNet18

ResNet18 is a residual neural network architecture consisting of skip connections that help prevent gradient vanishing problems. The model extracts hierarchical features from spectrogram images and performs binary classification.

3.7.2 DenseNet121

DenseNet121 uses dense connectivity between layers, allowing feature reuse and improved gradient flow. This architecture requires fewer parameters while maintaining strong feature extraction capability.

3.7.3 EfficientNet-B0

EfficientNet-B0 uses compound scaling to balance network depth, width, and resolution efficiently. It provides better accuracy with fewer parameters compared to traditional CNN architectures.

Diagram to Add:

- ResNet18 Architecture
- DenseNet121 Architecture
- EfficientNet-B0 Architecture

3.8 Optimization Techniques

Several advanced optimization techniques were applied to improve model performance under severe dataset imbalance conditions.

3.8.1 Focal Loss

Focal Loss reduces the influence of easy majority-class samples and forces the model to focus more on difficult anomaly samples.

The Focal Loss equation is:

$$FL(p_t) = -(1 - p_t)^\gamma \log(p_t)$$

where:

- p_t = predicted probability
- γ = focusing parameter

3.8.2 WeightedRandomSampler

WeightedRandomSampler increases the sampling probability of anomaly samples during training to balance the dataset distribution.

3.8.3 Data Augmentation

The following augmentation techniques were applied:

- Random Horizontal Flip
- Random Rotation
- Color Jitter
- Random Erasing
- Mixup Augmentation

3.8.4 Learning Rate Scheduling

OneCycleLR scheduling was used to improve convergence speed and optimize training performance.

Diagram to Add:

- Focal Loss Concept Diagram
- Data Augmentation Examples
- OneCycleLR Learning Rate Curve

3.9 Attention-Based Architecture (GAResNet)

To improve anomaly localization and interpretability, a custom attention-based architecture called GAResNet was developed.

The model integrates:

- Spatial Attention Module (SAM)
- Channel Attention Module (CAM)

Spatial attention identifies important regions in the spectrogram where anomalies occur, while channel attention determines the most relevant frequency components.

This allows the model to focus specifically on abnormal time-frequency regions instead of treating all pixels equally.

Diagram to Add:

- GAResNet Architecture Diagram
- Spatial Attention Mechanism
- Channel Attention Mechanism

3.10 Model Training

The models were trained using the PyTorch deep learning framework. Training was performed on GPU when available.

3.10.1 Training Configuration

Parameter	Value
Batch Size	32
Learning Rate	1×10^{-4}
Optimizer	Adam / AdamW
Epochs	5 – 20
Loss Function	CrossEntropy / Focal Loss
Scheduler	OneCycleLR

Table 3.3: Training Configuration

3.11 Performance Evaluation

The trained models were evaluated using several performance metrics.

3.11.1 Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

3.11.2 Confusion Matrix

The confusion matrix contains:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

These values help evaluate anomaly detection capability and class prediction behavior.
Diagram to Add:

- Confusion Matrix Diagram
- ROC Curve

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents the experimental results obtained from the proposed Smart Grid Anomaly Detection system. Multiple deep learning models were trained and evaluated using spectrogram images generated from smart grid time-series signals. The performance of baseline transfer learning models, optimized models, and custom attention-based architectures was analyzed using various evaluation metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix.

The primary objective of this study was to evaluate how effectively the models could distinguish between normal and anomalous grid conditions under severe class imbalance scenarios.

4.2 Experimental Setup

The experiments were conducted using Python and the PyTorch deep learning framework. Training was performed on GPU whenever available to improve computational speed.

4.2.1 Software and Hardware Configuration

Component	Specification
Programming Language	Python 3.x
Deep Learning Framework	PyTorch
Libraries Used	NumPy, Pandas, SciPy, Matplotlib
GPU Support	CUDA Enabled
Operating System	Windows
Image Size	224×224
Batch Size	32
Optimizer	Adam / AdamW

Table 4.1: Experimental Setup

4.3 Dataset Distribution

The smart grid dataset used in this study contains both normal and anomaly samples. The dataset was divided into training, validation, and testing sets using stratified splitting.

Dataset Category	Samples
Training Samples	545
Testing Samples	118
Normal Samples	456
Anomaly Samples	89
Normal Class Ratio	83.6%
Anomaly Class Ratio	16.4%

Table 4.2: Dataset Distribution

Diagram to Add:

- Pie Chart showing class imbalance
- Dataset Distribution Bar Graph

4.4 Baseline Model Results

Initially, baseline transfer learning models were trained using standard CrossEntropyLoss without advanced optimization techniques.

4.4.1 Baseline Performance Comparison

Model	Accuracy	F1-Score	ROC-AUC
ResNet18	45.76%	0.2558	0.4641
DenseNet121	26.27%	0.2810	0.5960
EfficientNet-B0	16.10%	0.2774	0.5502

Table 4.3: Baseline Model Performance

The baseline models struggled to detect anomalies effectively due to severe dataset imbalance. Most models became biased toward predicting the majority class (normal samples).

Diagram to Add:

- Baseline Accuracy Comparison Graph
- Baseline ROC Curve

4.5 Confusion Matrix Analysis

The confusion matrices revealed the prediction behavior of each model.

4.5.1 Baseline Confusion Matrix Results

Model	TP	TN	FP	FN
ResNet18	11	43	56	8
DenseNet121	17	14	85	2
EfficientNet-B0	19	0	99	0

Table 4.4: Baseline Confusion Matrix Results

The results show that EfficientNet-B0 predicted almost all samples as anomalies, while DenseNet121 produced a large number of false positives.

Diagram to Add:

- Confusion Matrix Heatmaps

4.6 Improved Model Results

Several advanced optimization techniques such as Focal Loss, WeightedRandomSampler, Mixup augmentation, label smoothing, and OneCycleLR scheduling were applied to improve model performance.

4.6.1 Improved Performance Comparison

Model	Baseline Accuracy	Improved Accuracy	Improvement
ResNet18	45.76%	83.05%	+37.29%
DenseNet121	26.27%	83.05%	+56.78%
EfficientNet-B0	16.10%	83.90%	+67.80%

Table 4.5: Improved Model Performance

The optimized models achieved significantly higher classification accuracy after applying advanced deep learning optimization techniques.

Diagram to Add:

- Accuracy Improvement Bar Chart
- Baseline vs Improved Comparison Graph

4.7 Training Dynamics Analysis

The training loss and validation accuracy curves showed improved convergence behavior after optimization.

4.7.1 Training Observations

- OneCycleLR scheduling improved convergence speed.
- Mixup augmentation improved model robustness.

- Label smoothing reduced overconfidence in predictions.
- Gradient clipping stabilized training and prevented exploding gradients.
- WeightedRandomSampler improved anomaly sample exposure during training.

Diagram to Add:

- Training Loss Curve
- Validation Accuracy Curve
- Learning Rate Scheduler Curve

4.8 GAResNet Performance

A custom attention-based architecture called GAResNet was developed to improve anomaly localization and interpretability.

4.8.1 GAResNet Results

Model	Accuracy	Recall	F1-Score
GAResNet	33.05%	78.95%	0.28

Table 4.6: GAResNet Performance

Although GAResNet achieved lower overall accuracy, it provided significantly higher anomaly recall. This makes the architecture highly suitable for smart grid monitoring systems where detecting anomalies is more important than maximizing overall accuracy.

Diagram to Add:

- GAResNet Architecture Diagram
- Attention Map Visualization

4.9 Comparison of All Models

Model	Accuracy	Recall	F1-Score
ResNet18 Baseline	45.76%	16.60%	0.17
EfficientNet-B0	83.90%	0.00%	0.00
GAResNet	33.05%	78.95%	0.28
Attentional ResNet-50	83.90%	10.53%	0.16

Table 4.7: Overall Model Comparison

The results clearly show that high accuracy alone is not sufficient for anomaly detection tasks involving imbalanced datasets. Models achieving high accuracy often fail to detect anomalies effectively.

GAResNet achieved the highest recall, making it more suitable for real-world fault-sensitive smart grid applications.

Diagram to Add:

- Overall Performance Comparison Graph
- Accuracy vs Recall Scatter Plot

4.10 Discussion

The experimental results demonstrate that severe class imbalance is the major challenge in smart grid anomaly detection. Traditional transfer learning models achieved high accuracy by predicting most samples as normal. However, these models failed to detect rare anomaly samples effectively.

Advanced optimization techniques improved overall classification performance significantly. Techniques such as Focal Loss and WeightedRandomSampler helped reduce class imbalance effects, while OneCycleLR scheduling and Mixup augmentation improved convergence and generalization.

The custom attention-based GAResNet architecture successfully improved anomaly recall by focusing on important time-frequency regions within spectrogram images. This demonstrates that attention mechanisms are highly effective for smart grid anomaly localization and interpretable AI systems.

4.11 Chapter Summary

This chapter presented the experimental results and performance analysis of the proposed Smart Grid Anomaly Detection system. Baseline transfer learning models, optimized models, and custom attention-based architectures were evaluated using multiple performance metrics. The results demonstrated that attention-based models provide superior anomaly sensitivity and interpretability compared to standard CNN architectures. The next chapter presents the conclusion and future scope of the project.

Conclusions

This study presented a Smart Grid Anomaly Detection system using spectrogram-based deep learning techniques for monitoring power grid abnormalities. Time-series smart grid signals were converted into spectrogram images using STFT, allowing deep learning models to analyze frequency and time-domain patterns effectively.

Several transfer learning models such as ResNet18, DenseNet121, and EfficientNet-B0 were evaluated for anomaly classification. Advanced optimization techniques including Focal Loss, WeightedRandomSampler, Mixup augmentation, and OneCycleLR scheduling significantly improved overall model performance. In addition, custom attention-based architectures such as GAResNet improved anomaly sensitivity and interpretability by focusing on important time-frequency regions in the spectrograms.

The experimental results demonstrated that attention-based deep learning approaches are effective for smart grid anomaly detection, especially in highly imbalanced datasets. The proposed system can support real-time monitoring and early fault detection in smart grid infrastructure.

4.12 Future Scope

Future work can focus on collecting larger and more balanced datasets to improve anomaly detection performance further. Transformer-based architectures and hybrid CNN-LSTM models can also be explored for better temporal feature extraction. Real-time deployment using IoT-enabled smart grid devices and edge computing platforms can enhance practical implementation. Additional explainable AI techniques may also be integrated to improve model transparency and operator trust.

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